

Transmeta Fights back with Astro

In answer to the Pentium-M, Transmeta is preparing a completely redesigned architecture and new bus technology for the TM8000-code-named the Astro. The company released its first chip, the low-power Crusoe designed for ultra-portable notebooks and Tablet PCs, in January 2000. It uses code-morphing software that both interprets x86 instructions from applications, and also translates frequently used x86 instructions into the Crusoe's VLIW (Very Long Instruction Word) instructions. This reduces the number of transistors the chip needs, moving the complexity of the processor into software, which saves power, according to Transmeta.

Transmeta also included three new bus technologies in the forthcoming Astro processor. The company will use Hypertransport, a point-to-point interconnect technology, for the processor's front-side bus. It will support DDR400 (double data rate at 400 MHz) memory, but will maintain backward compatibility with the more widely-used DDR266 and DDR333 memory standards. The chip will support AGP 4X for very high-speed graphics for Windows applications.

802.11a standard goes up to 54 Mbps but operates in the 5.0 GHz range. 802.11g matches the speed of 802.11a (54 Mbps), but uses the 802.11b (2.4 GHz) frequency. This means that a single 802.11g radio should be able to support the choice of 11 Mbps or 54 Mbps speeds.

Analysts, however, aver that if you want notebooks to connect to 802.11a or 802.11g networks (and not only to WiFi), Centrino-based systems won't play unless the built-in radio is disabled and a third-party adapter is added.

Competition beware

The Pentium-M notebooks would run for about 4-5 hours without recharging. The battery life is not drained only by the CPU, but also by the choice of other components such as displays, hard drives and memory. Hence, expecting the notebooks to run continuously for 8-10 hours may not be realistic, at least for another couple of years. But this does not rob from the achievement that is Centrino.

One would tend to agree with Avtar Saini, Director of the South Asian endeavours of Intel Asia Electronics, who says: "This breakthrough innovation, together with industry-wide investment and WiFi hotspot deployments, brings computing and communications capabilities to businesses and consumers, adding new value to mobile PCs." Intel has already been working with leading wireless network service providers, hotels, airports, and retail and restaurant chains worldwide to accelerate deployment and increase awareness of wireless public hotspots. One needs to remember that not all notebook makers favour Intel's implementation of 802.11b technology, as pointed out earlier.

Computer shoppers can identify the notebooks by the Centrino brand logo in a striking magenta colour and the 'Intel Inside' mark. If Centrino becomes a success, the competition will have to pull its pants up. They will have to do much more than merely modify their desktop processor solutions. 

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